

Artificial Intelligence

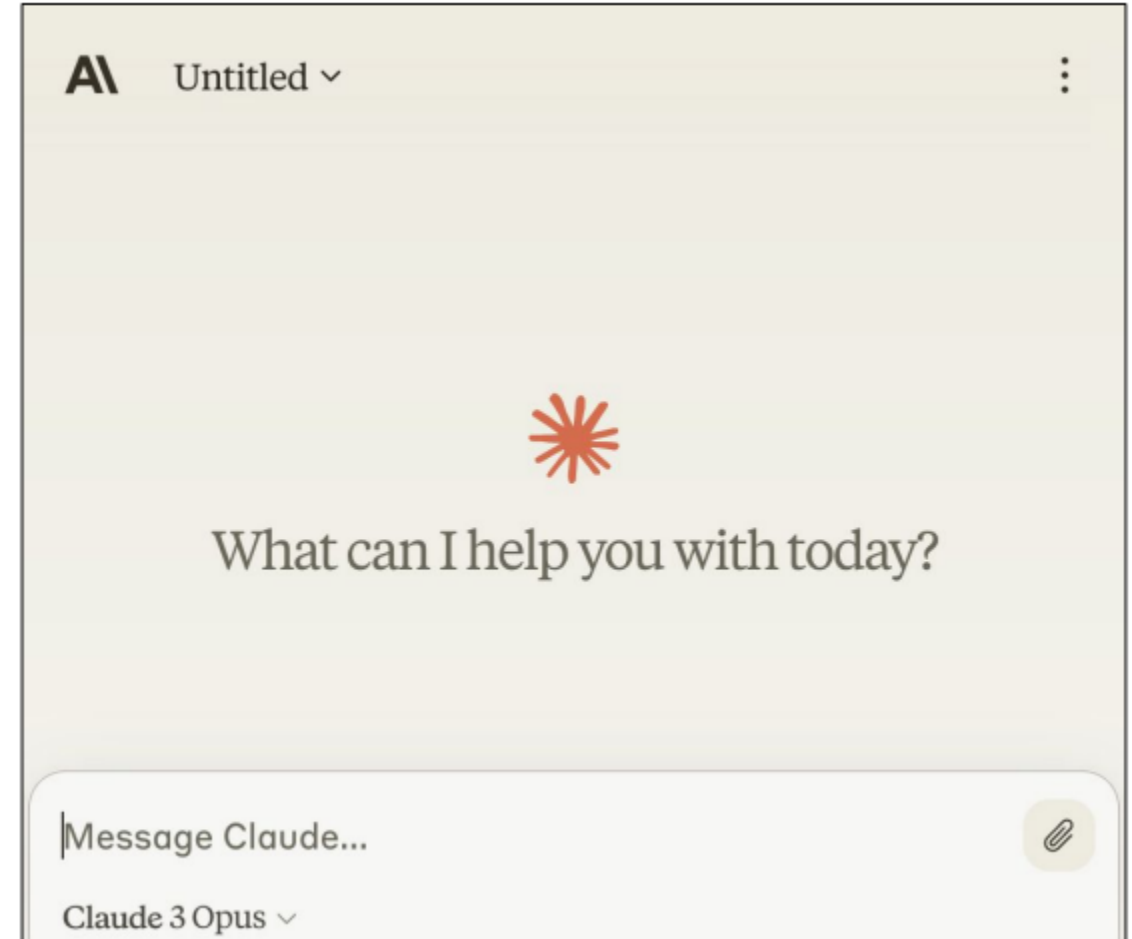
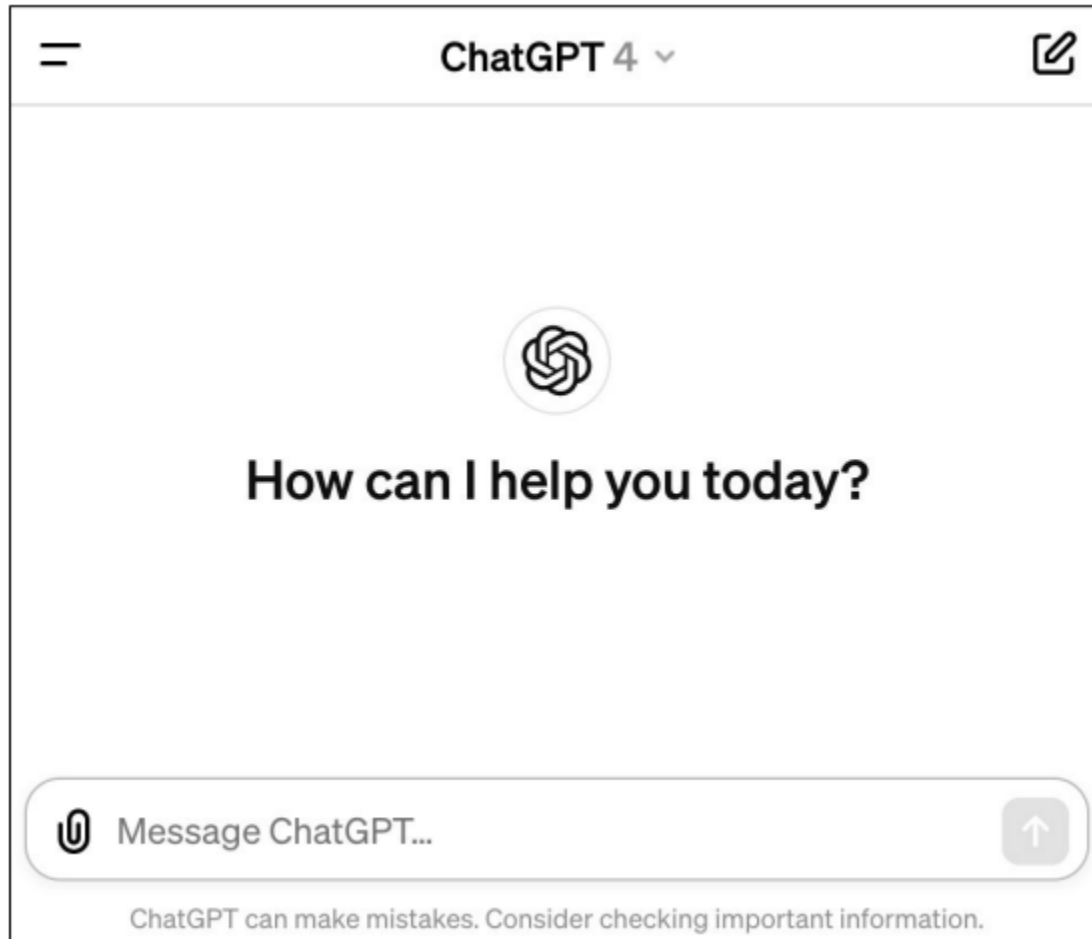
Lecture10 – LLMs & Transformers



Feed Forward Layers



Today's IA

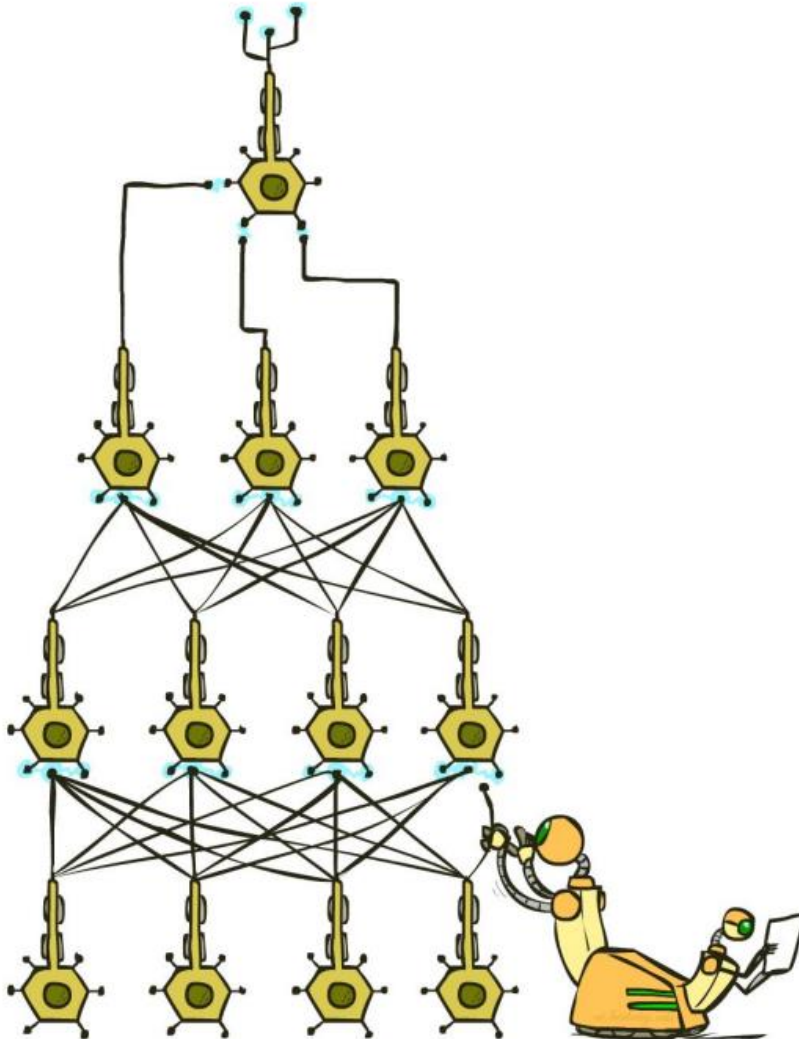


Large Language Models

- Feature engineering
 - Text tokenization
 - Word embeddings
- Deep neural networks
 - Autoregressive models
 - Self-attention mechanisms
 - Transformer architecture
- Multi-class classification
- Supervised learning
 - Self-supervised learning
 - Instruction tuning
- Reinforcement learning
 - ... from human feedback (RLHF)



Deep Neural Networks



- Input: some text
 - “The dog chased the”
- Output: more text
 - ... “ ball”
- Implementation:
 - Linear algebra
 - How??



Text Tokenization

GPT-3.5 & GPT-4

GPT-3 (Legacy)

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🖐

Sequences of characters commonly found next to each other may be grouped together: 1234567890

Clear

Show example

Tokens

57

Characters

252



Text Tokenization

GPT-3.5 & GPT-4

GPT-3 (Legacy)

Many words map to one token, but some don't: indivisible.

Unicode characters like emojis may be split into many tokens containing the underlying bytes: 🍌🍌🍌🍌🍌

Sequences of characters commonly found next to each other may be grouped together: 1234567890

Text

Token IDs

Tokens

57

Characters

252

<https://gptforwork.com/tools/tokenizer>



Text Tokenization

GPT-3.5 & GPT-4 GPT-3 (Legacy)

```
[8607, 4339, 2472, 311, 832, 4037, 11, 719, 1063, 1541, 956, 25, 3687,  
23936, 382, 35020, 5885, 1093, 100166, 1253, 387, 6859, 1139, 1690,  
11460, 8649, 279, 16940, 5943, 25, 11410, 97, 248, 9468, 237, 122, 271,  
1542, 45045, 315, 5885, 17037, 1766, 1828, 311, 1855, 1023, 1253, 387,  
41141, 3871, 25, 220, 4513, 10961, 16474, 15]
```

Text

Token IDs

Tokens

57

Characters

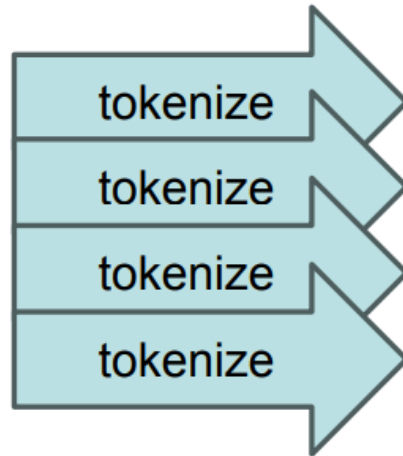
252



Word Embeddings

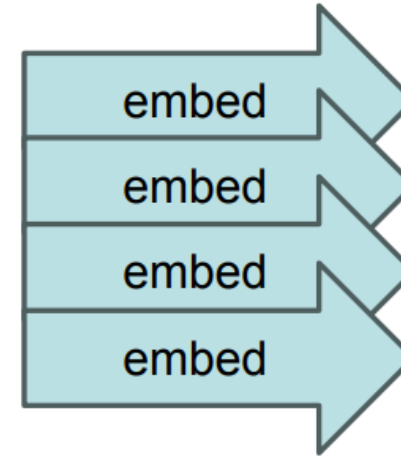
- Input: some text

- “The”
- “ dog”
- “ chased”
- “ the”



one-hot

[791]
[5679]
[62920]
[279]

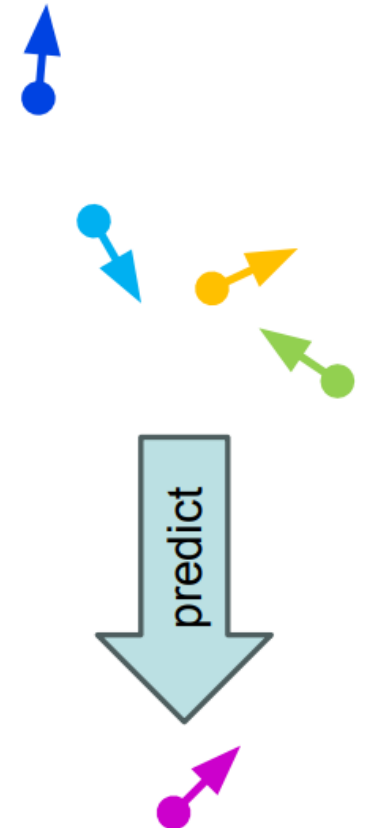
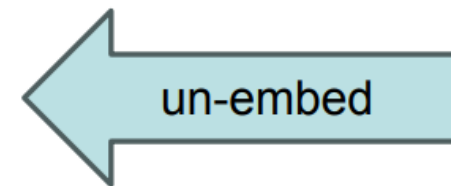


- Output: more text

- “ ball”

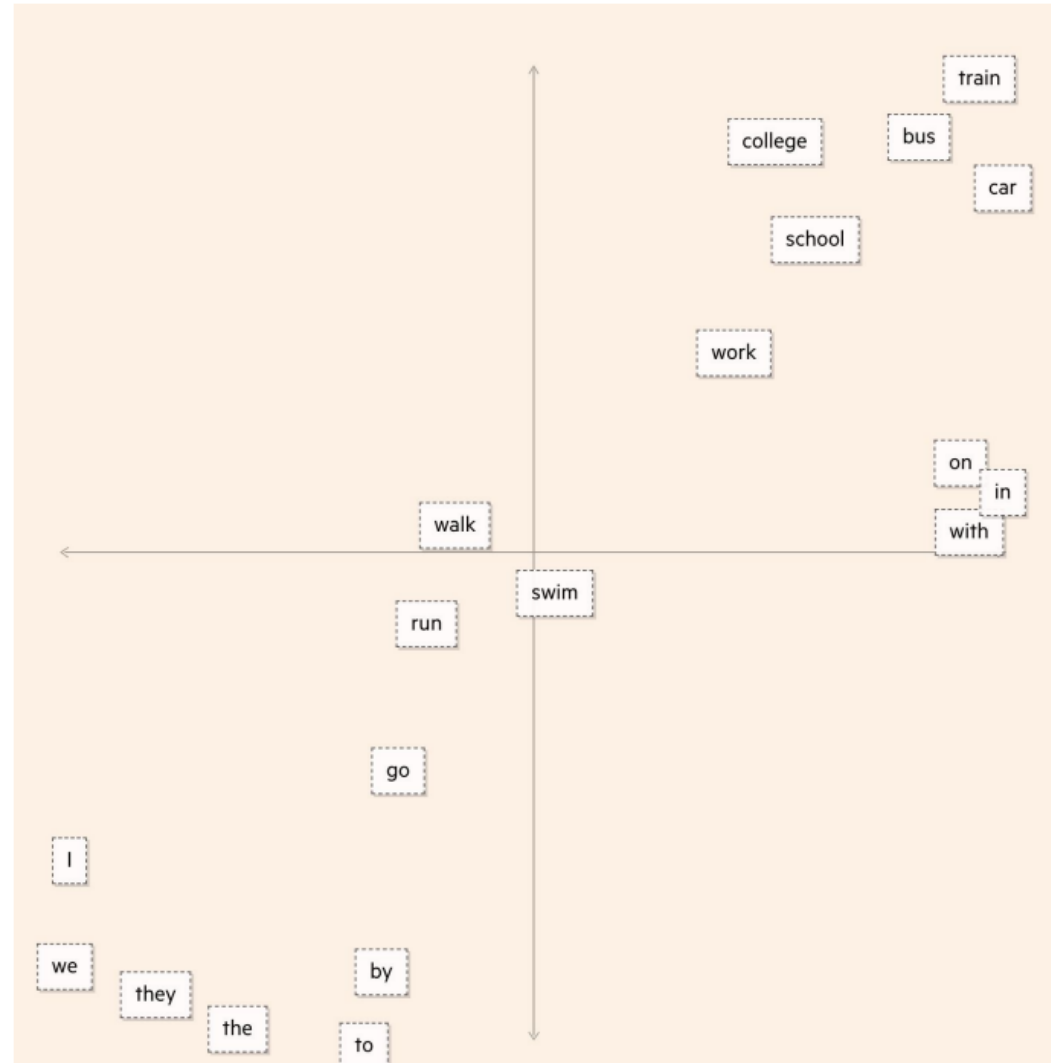


[5041]



What do Word embeddings look like?

- Words cluster by similarity:

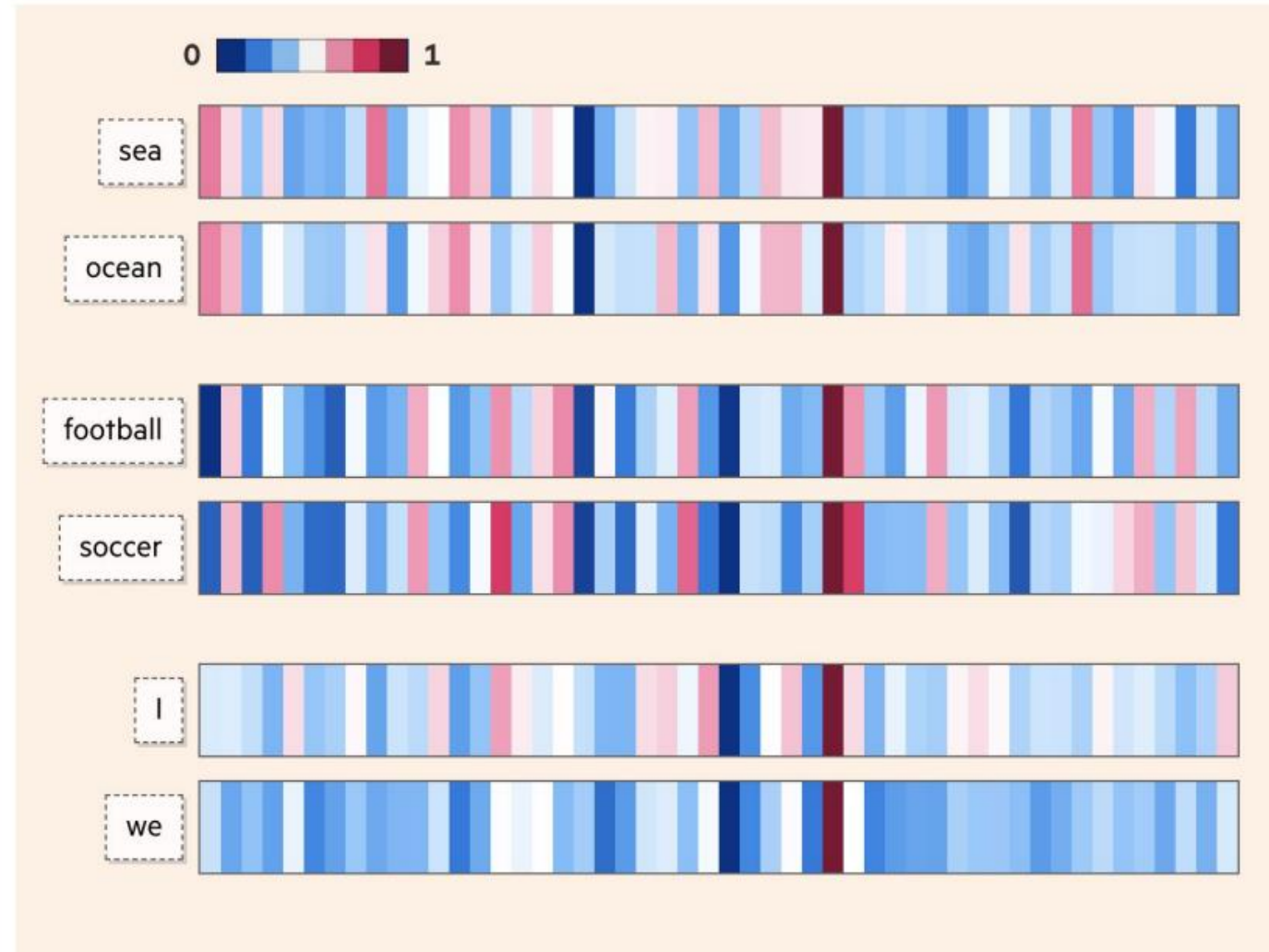


<https://ig.ft.com/generative-ai/>
<https://projector.tensorflow.org/>



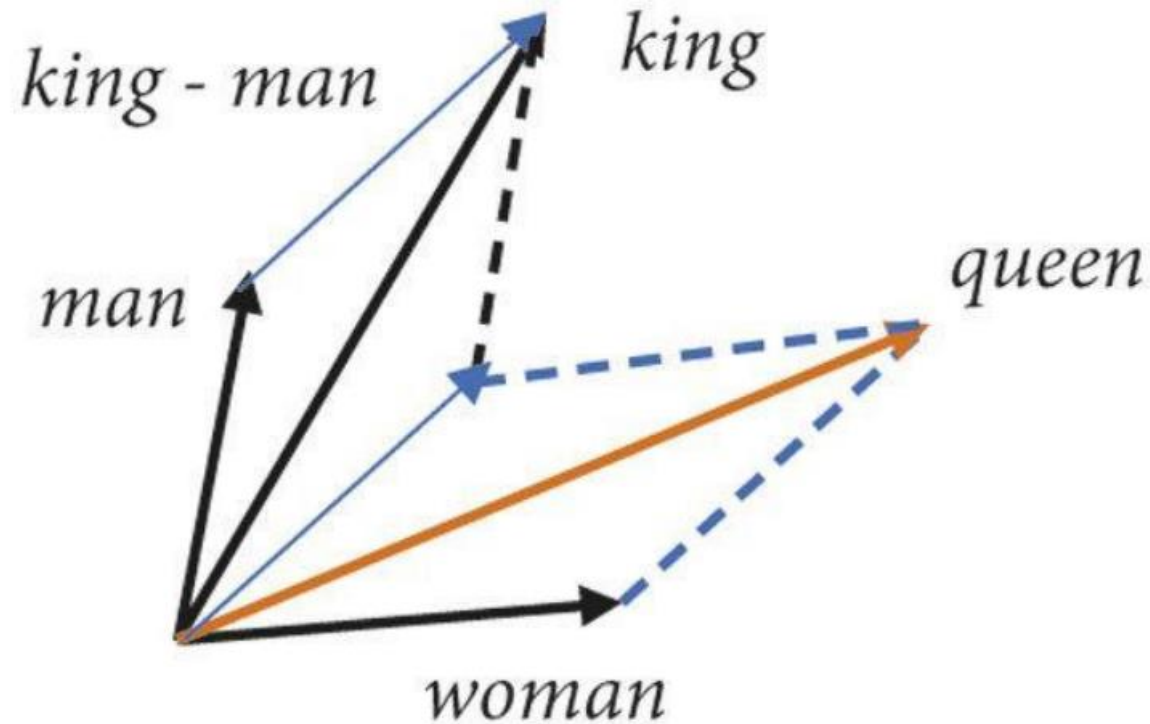
What do Word embeddings look like?

- Features learned in language models:



What do Word embeddings look like?

- Signs of sensible algebra in embedding space:



$$\text{king} - \text{man} + \text{woman} \approx \text{queen}$$

[Efficient estimation of word representations in vector space, Mikolov et al, 2013]



What do Word embeddings look like?

Man is to Computer Programmer as Woman is to Homemaker? Debiasing Word Embeddings

Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, Adam Kalai



Interactive explainer of modern language models

Artificial Intelligence

Generative AI exists because of the transformer

This is how it works

By Visual Storytelling Team and Madhumita Murgia in London SEPTEMBER 11 2023

<https://ig.ft.com/generative-ai/>

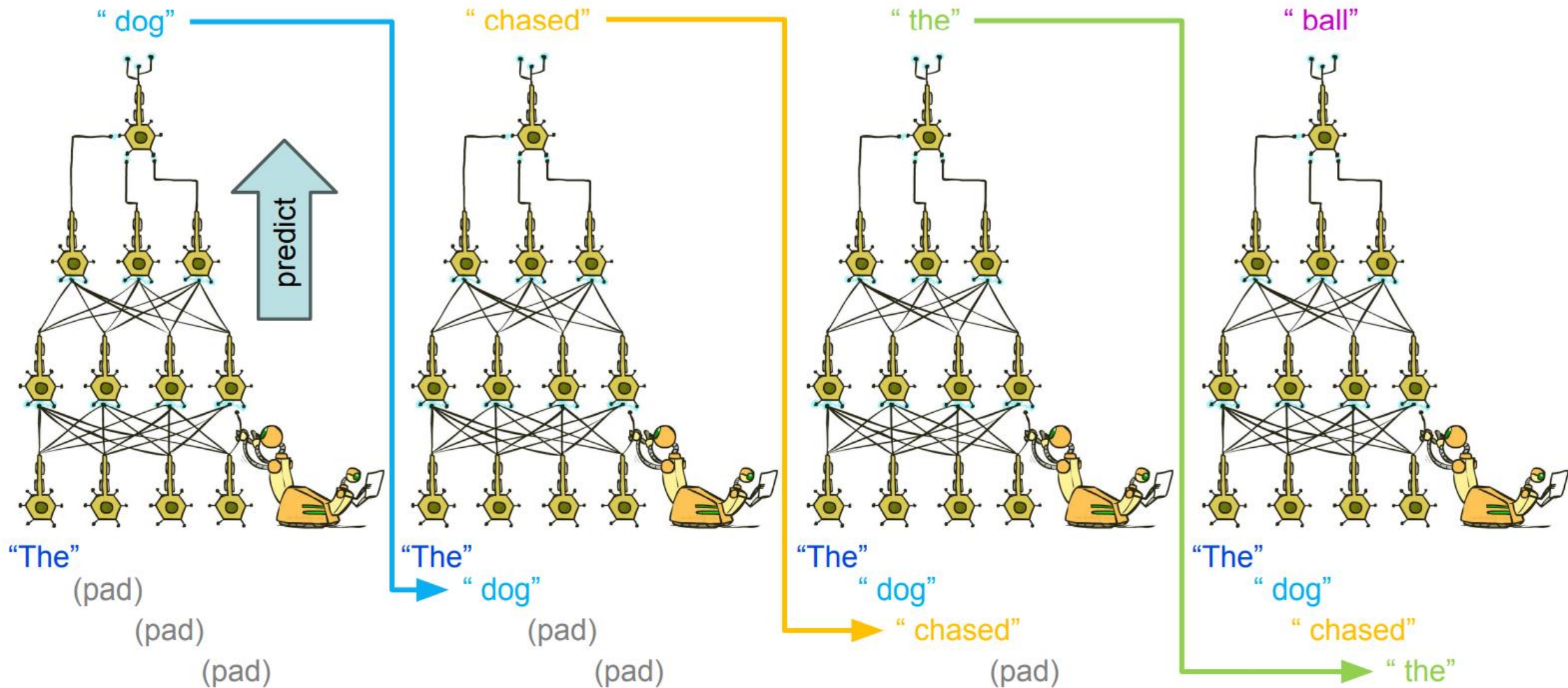


Large Language Models

- ~~Feature engineering~~
 - ~~Text tokenization~~
 - ~~Word embeddings~~
- Deep neural networks
 - Autoregressive models
 - Self-attention mechanisms
 - Transformer architectures
- Multi-class classification
- Supervised learning
 - Self-supervised learning
 - Instruction tuning
- Reinforcement learning
 - ... from human feedback (RLHF)

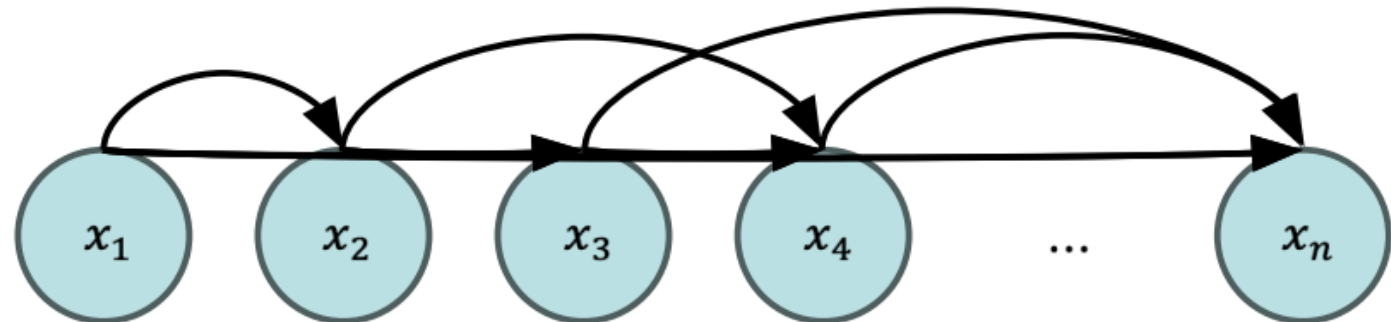


Autoregressive Models

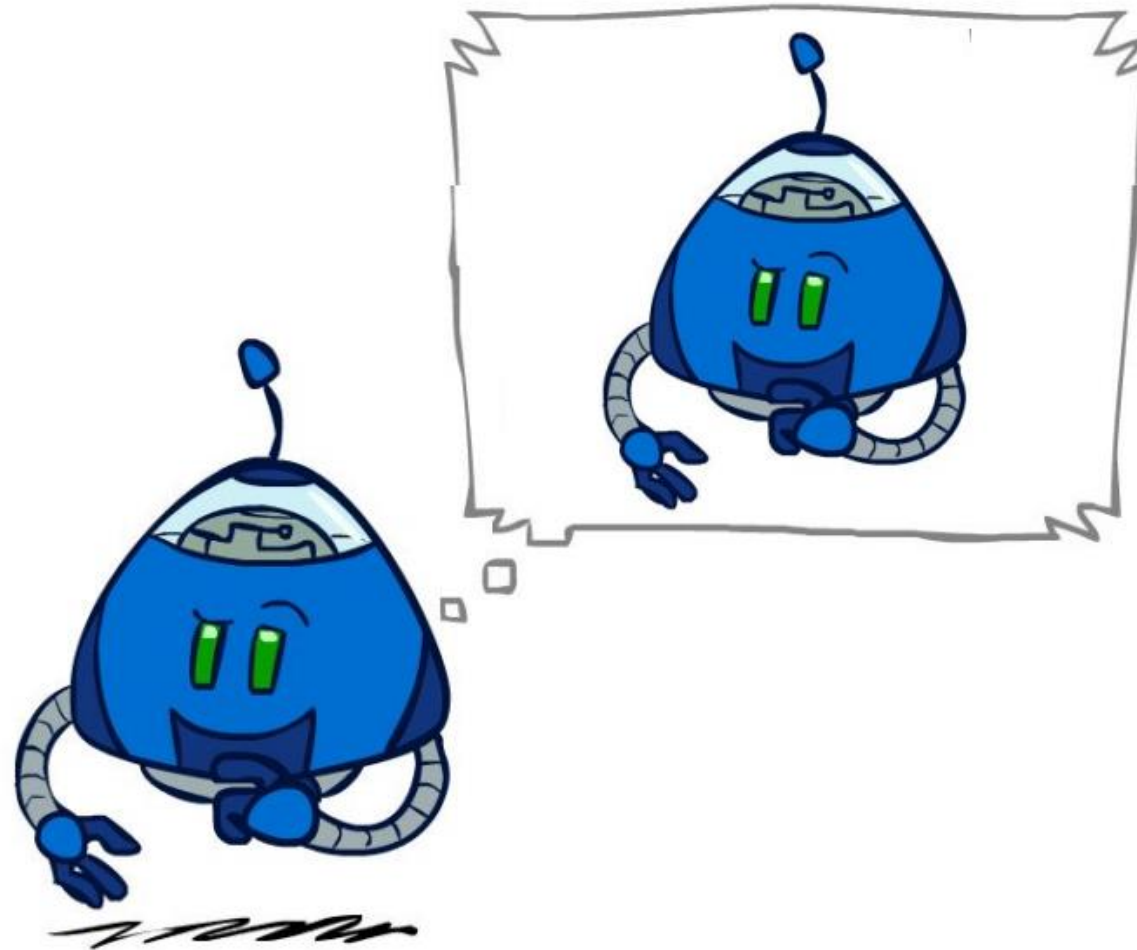


Autoregressive Models

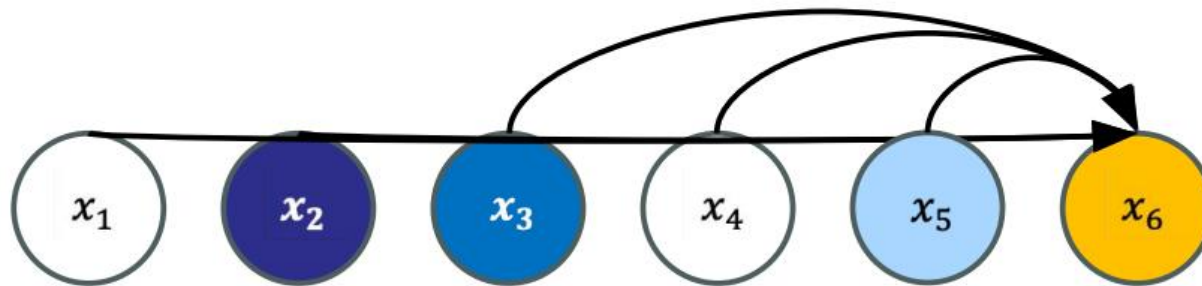
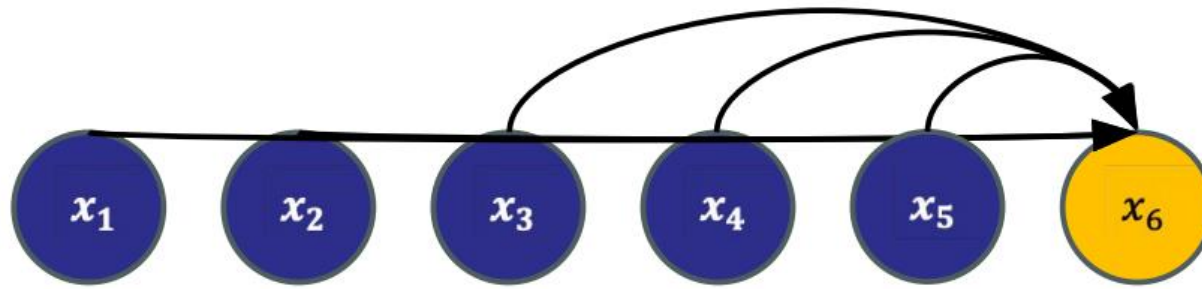
- Predict output one piece at a time (e.g. word, token, pixel, etc.)
- Concatenate: input + output
- Feed result back in as new input
- Repeat



Self-Attention Mechanisms



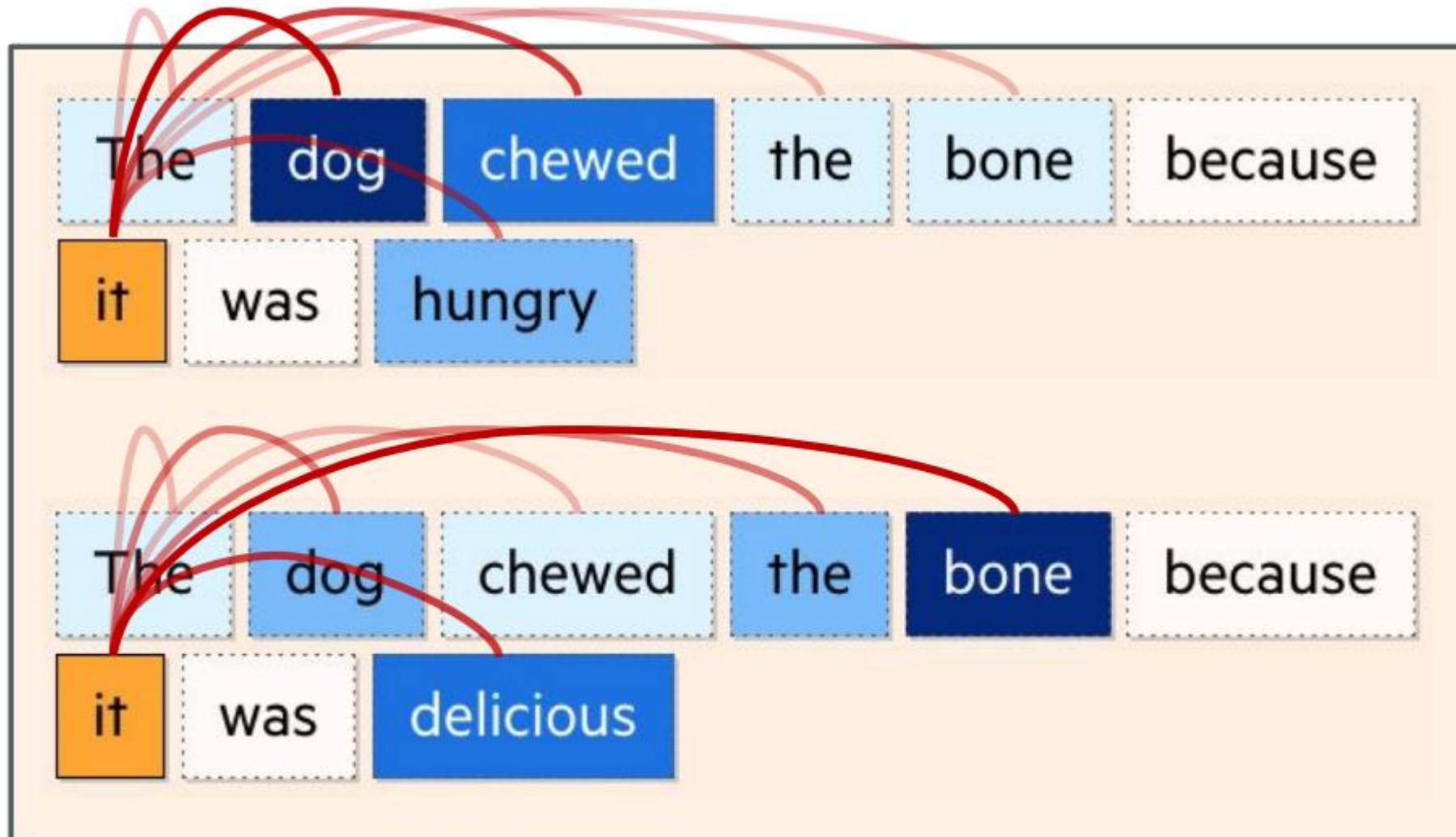
Self-Attention Mechanisms



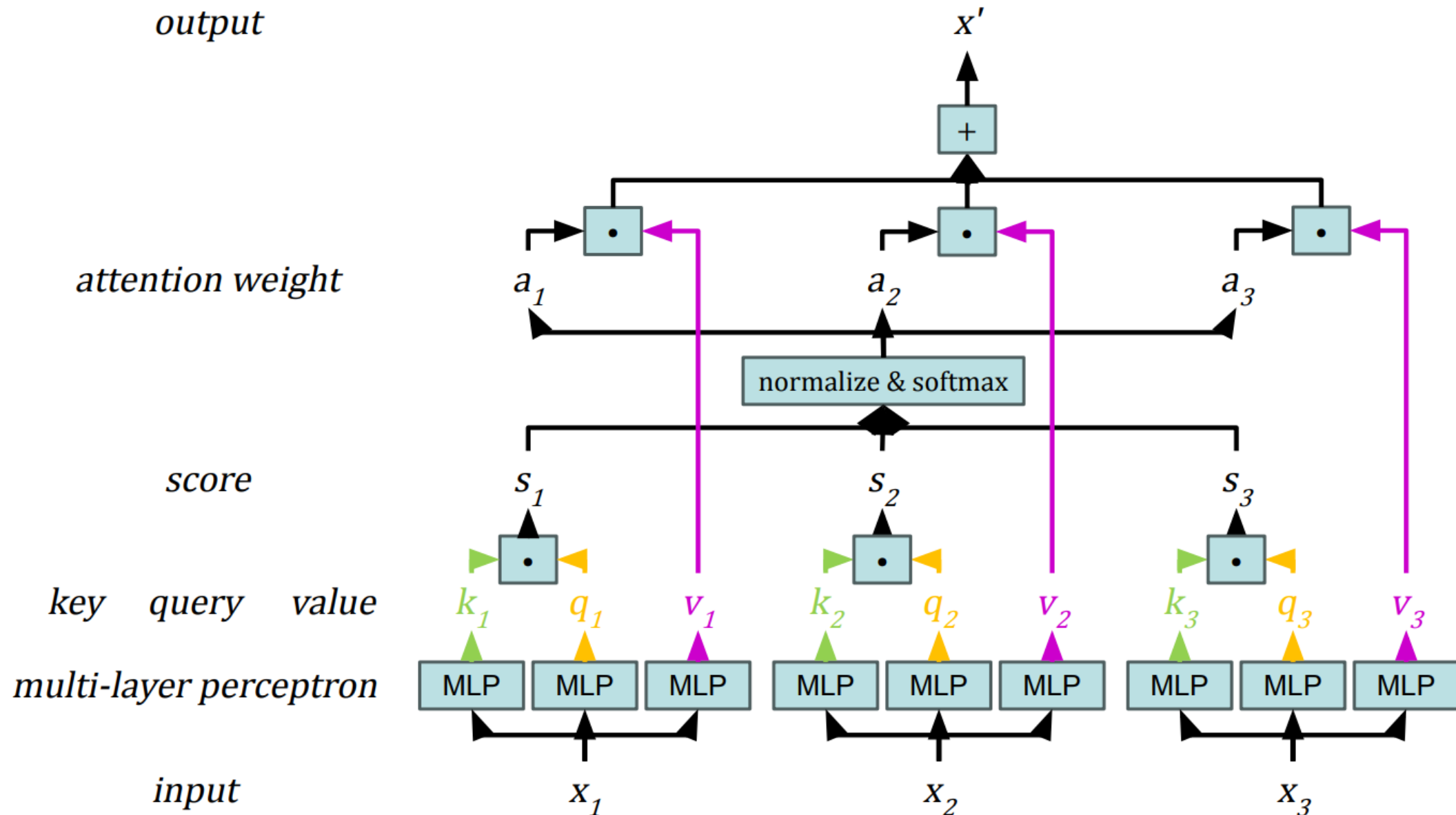
- Instead of conditioning on *all* input tokens equally...
- Pay more attention to relevant tokens!



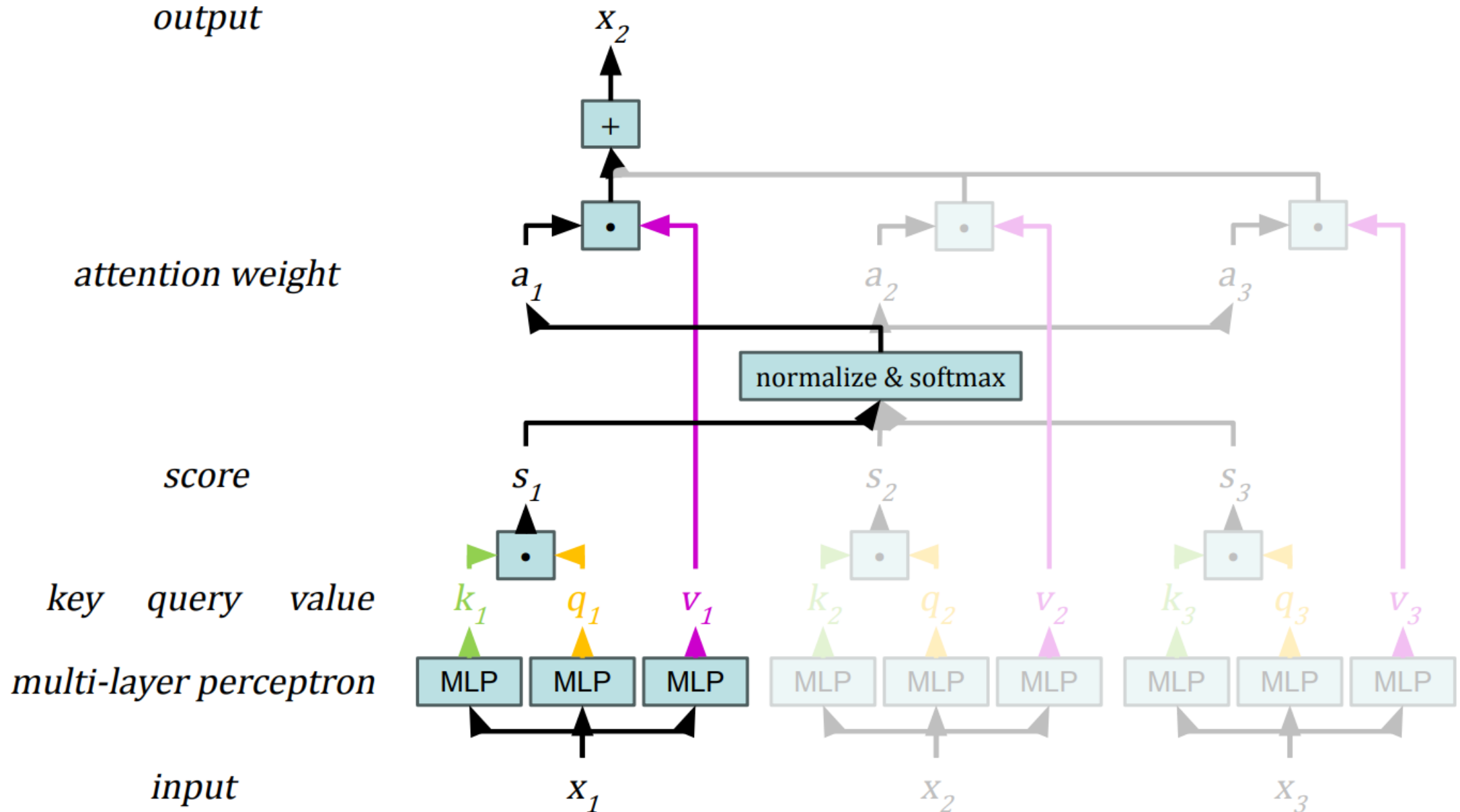
Self-Attention Mechanisms



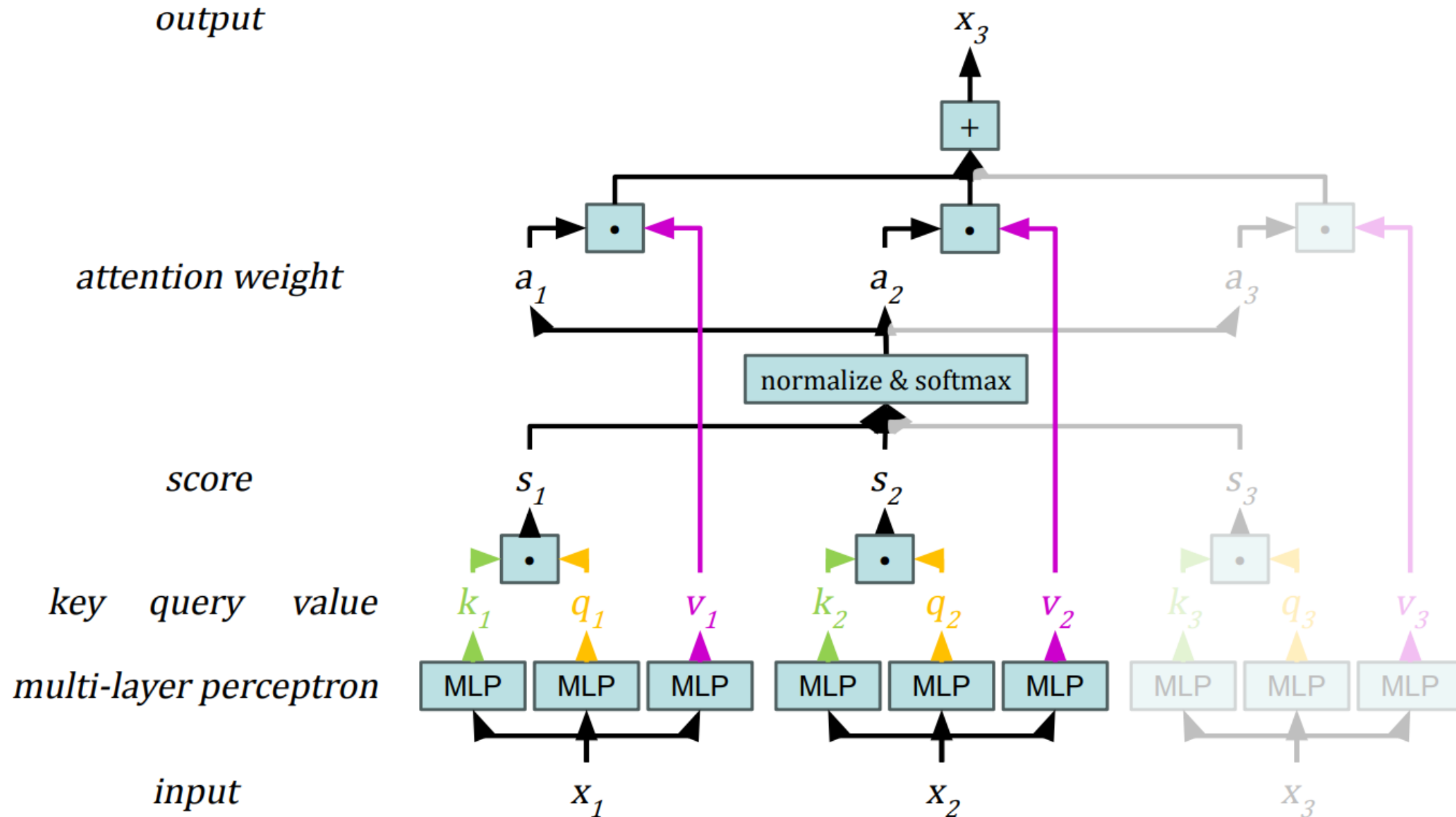
Self-Attention Mechanisms



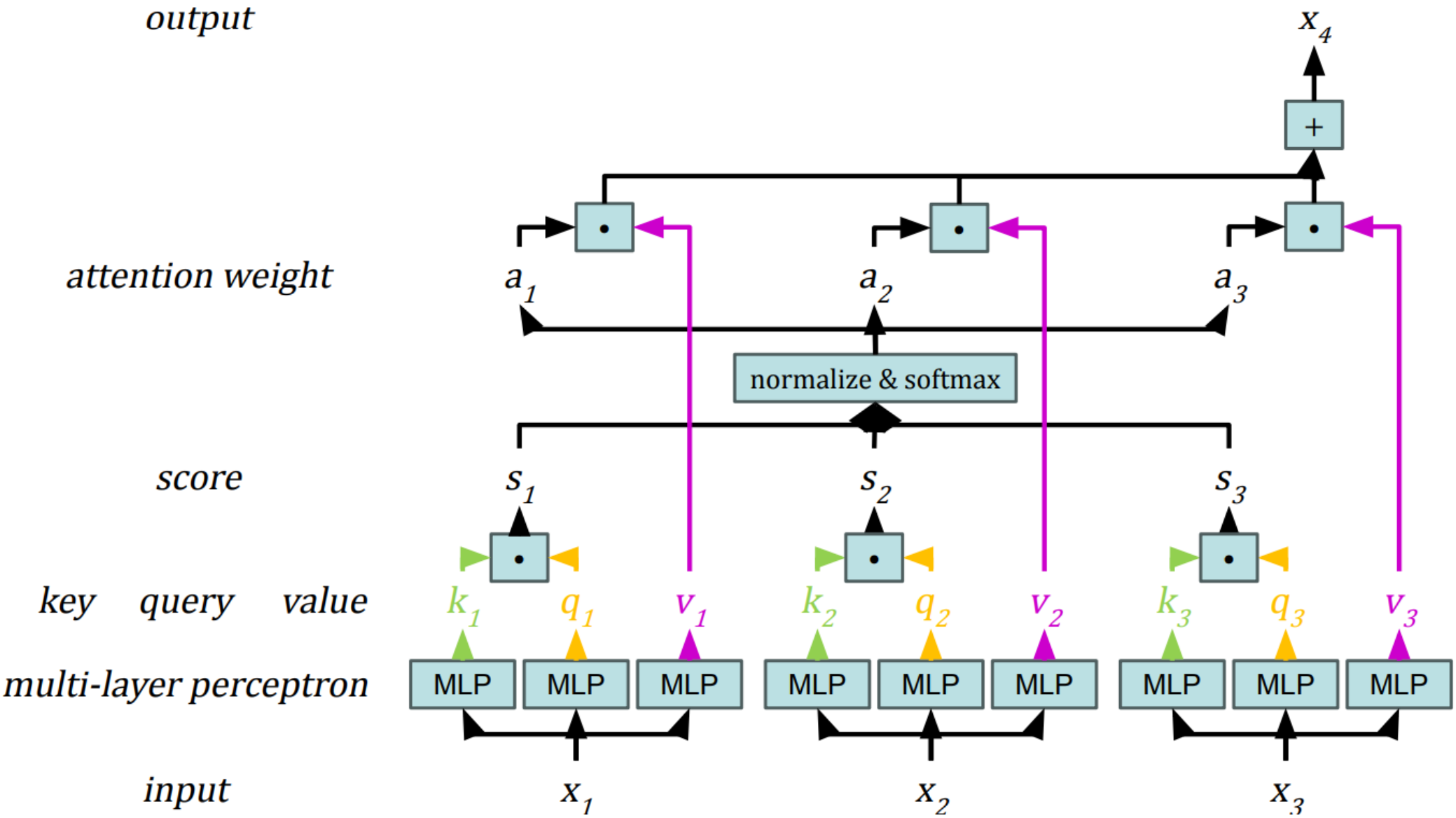
Self-Attention Mechanisms



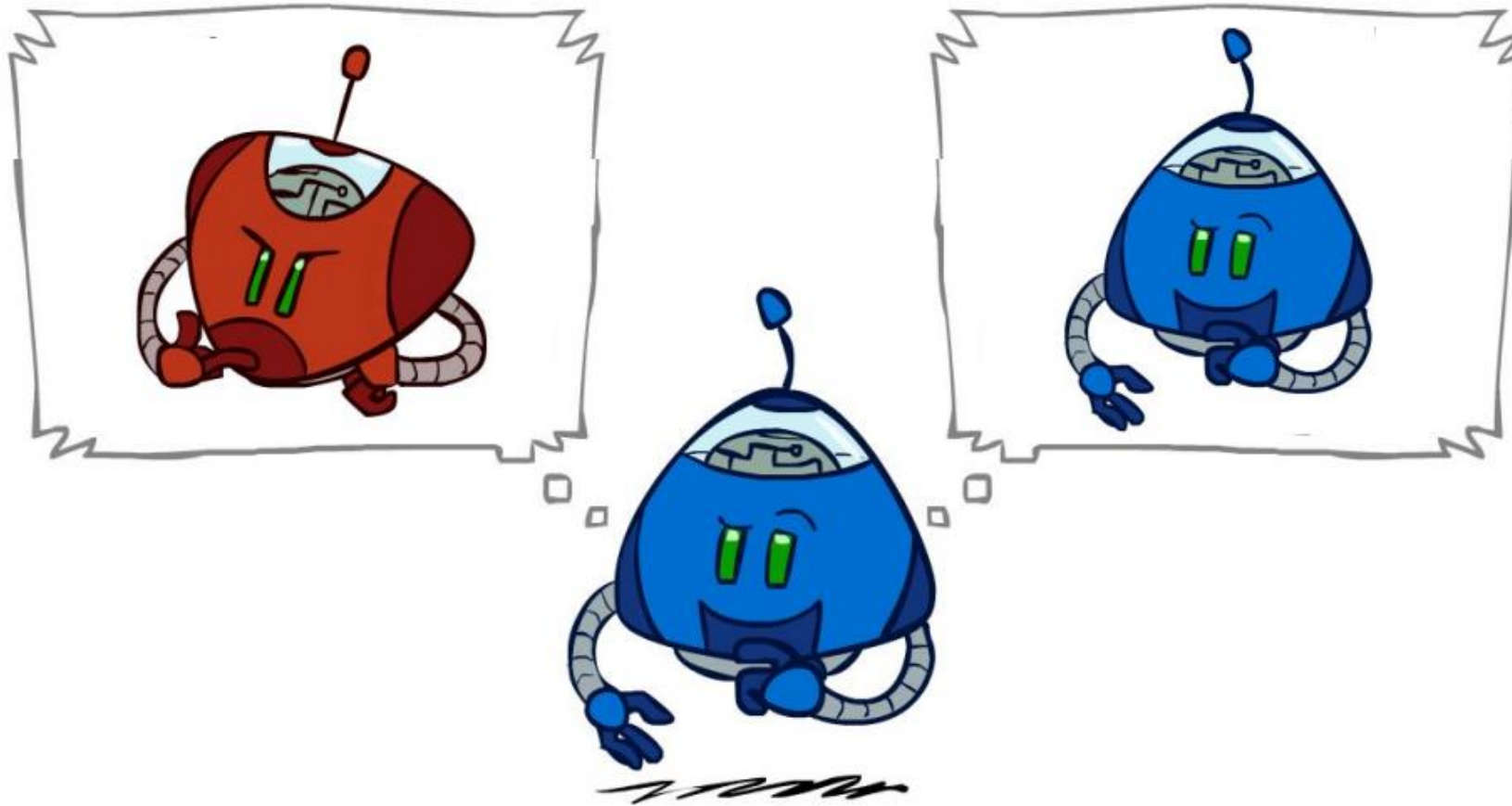
Self-Attention Mechanisms



Self-Attention Mechanisms

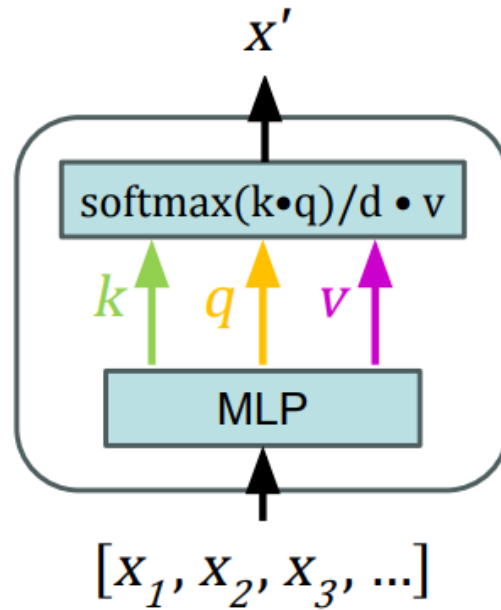


Multi-Headed Attention

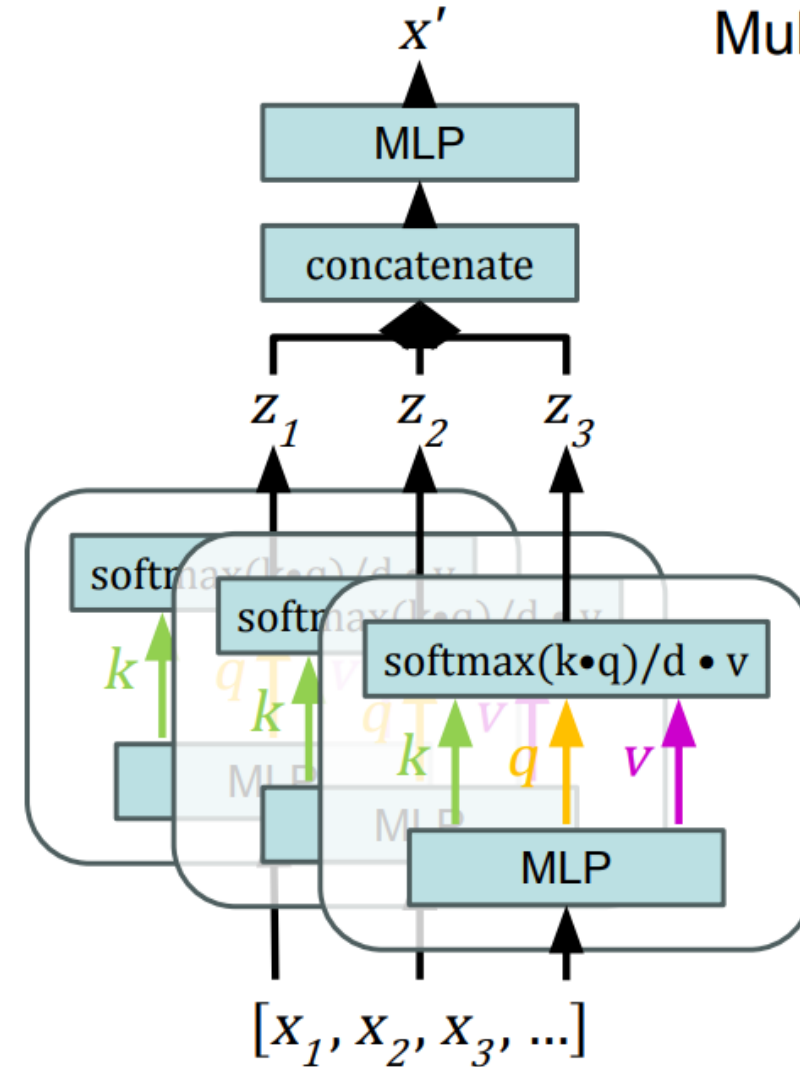


Multi-Headed Attention

Single-headed

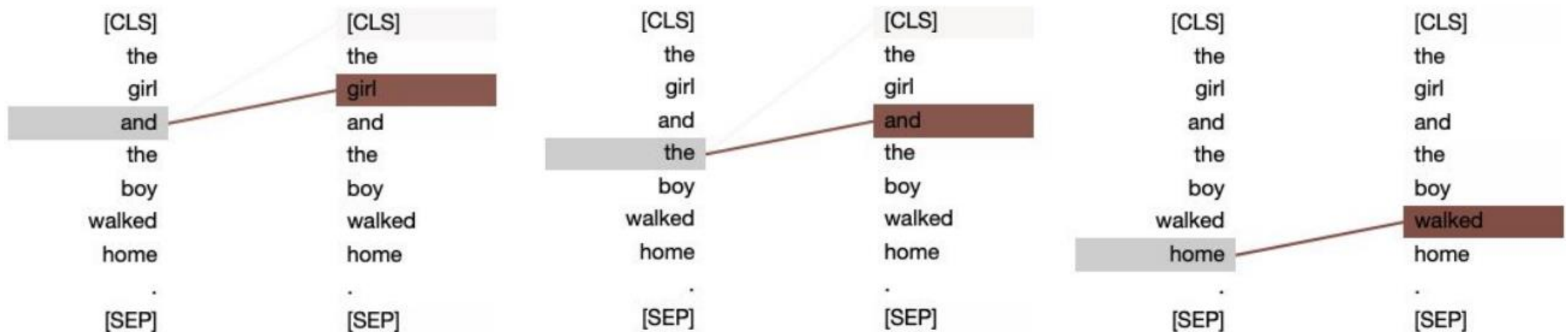


Multi-headed



Multi-Headed Attention

Head 6: previous word

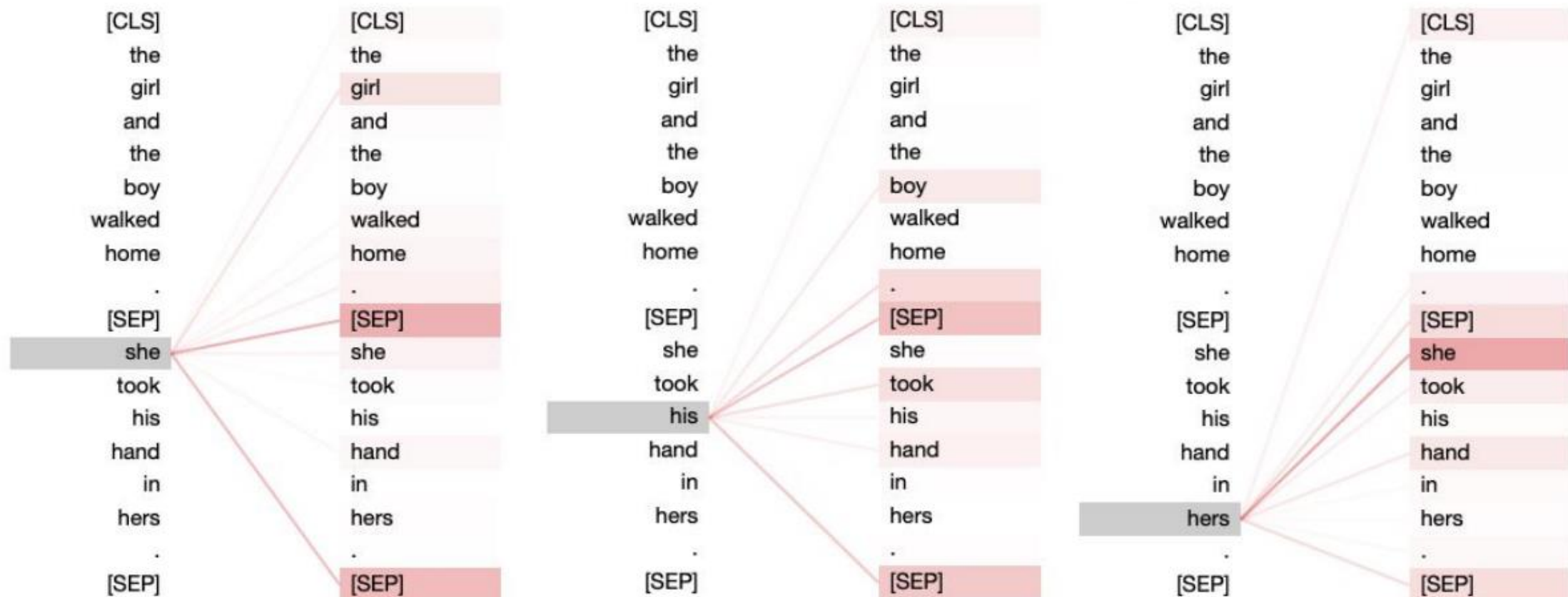


“The girl and the boy walked home”



Multi-Headed Attention

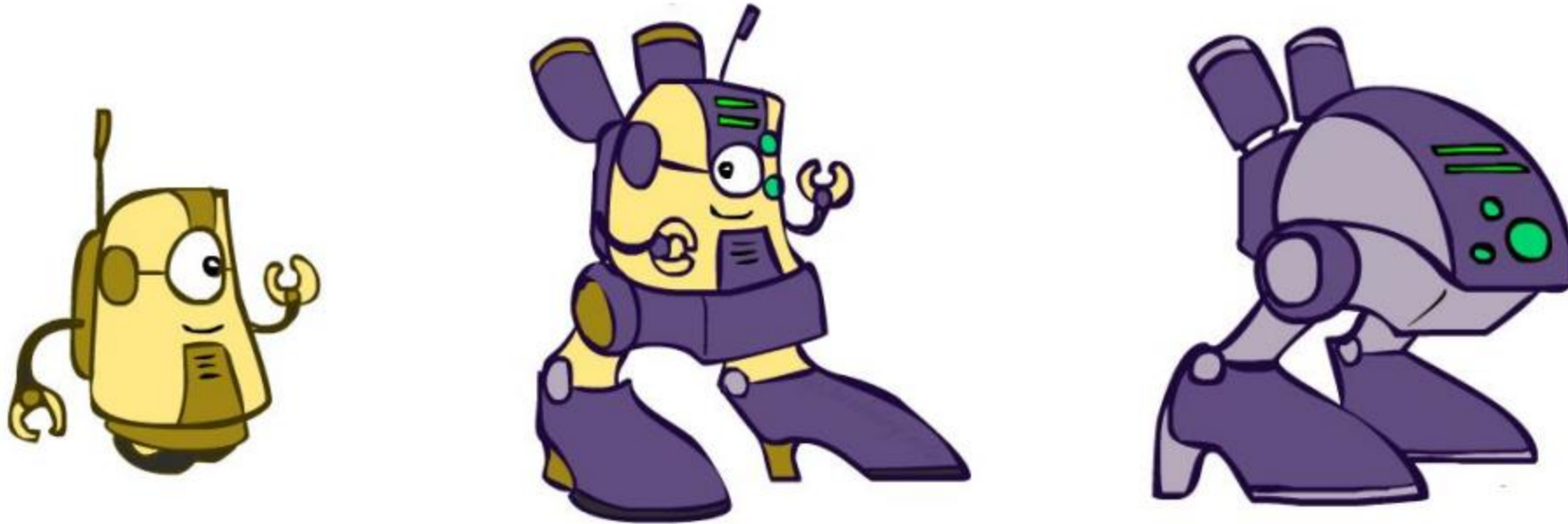
Head 4: pronoun references



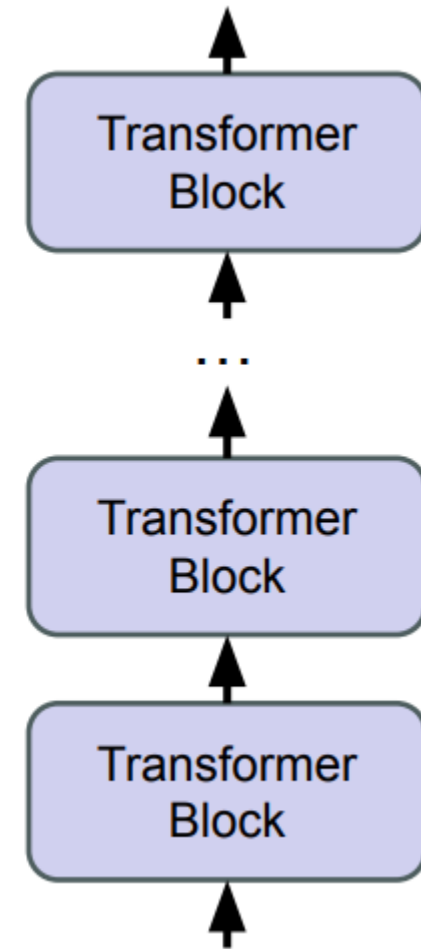
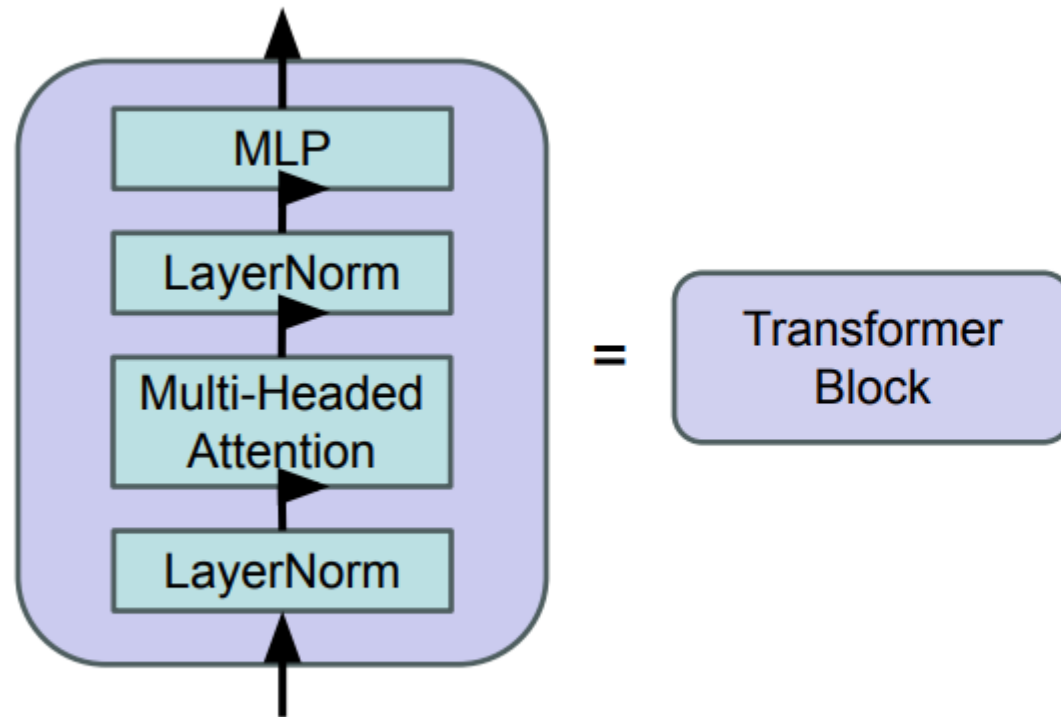
“She took his hand in hers”



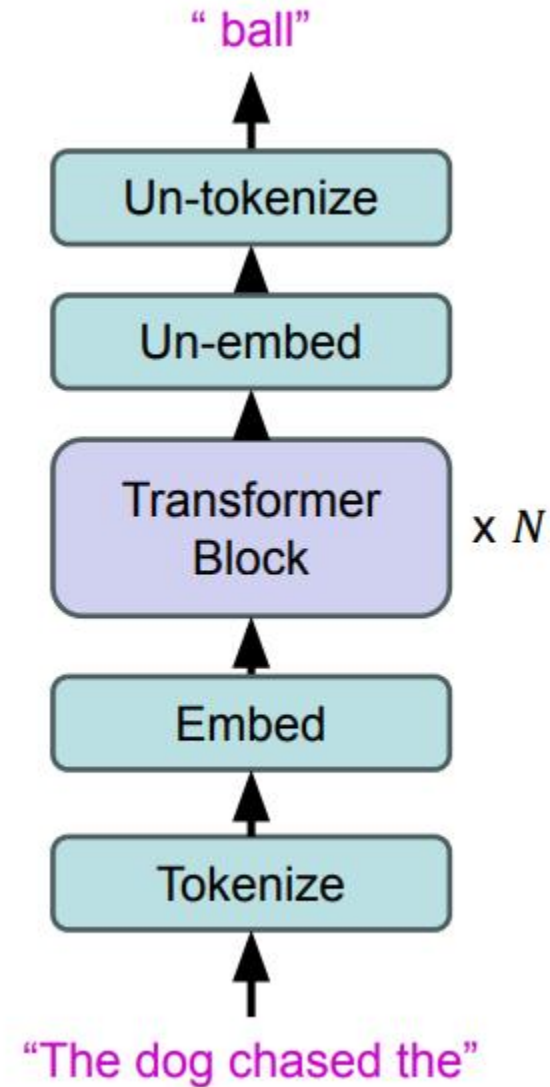
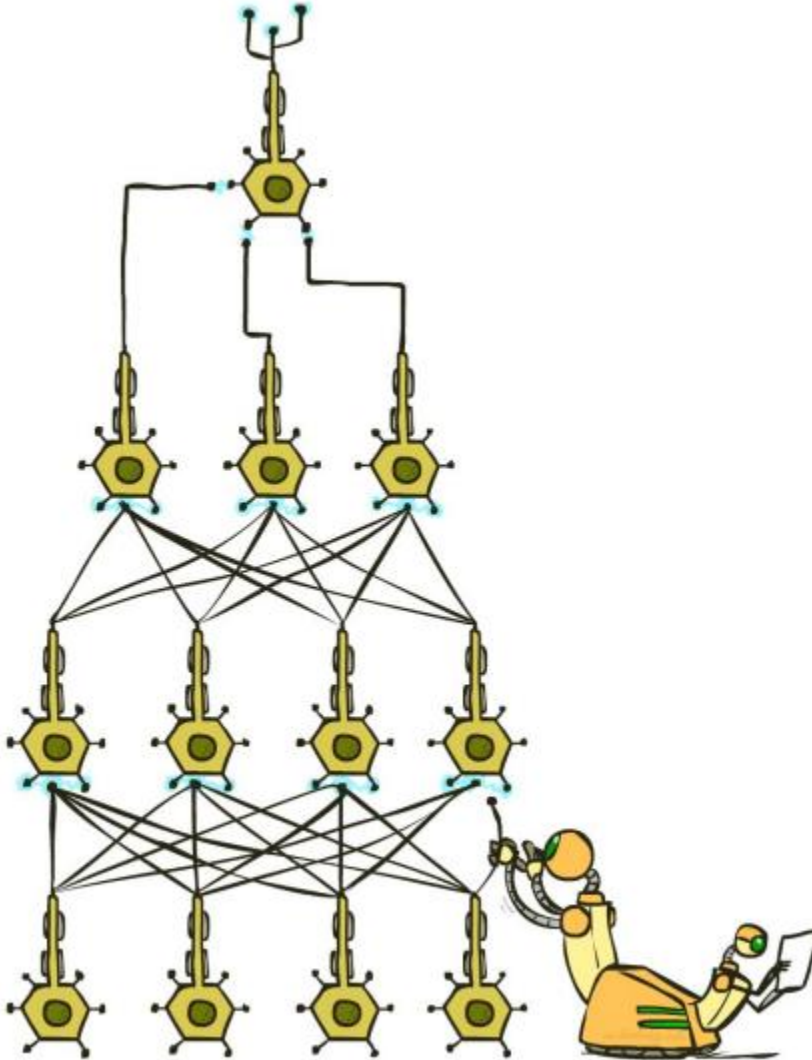
Transformer Architecture



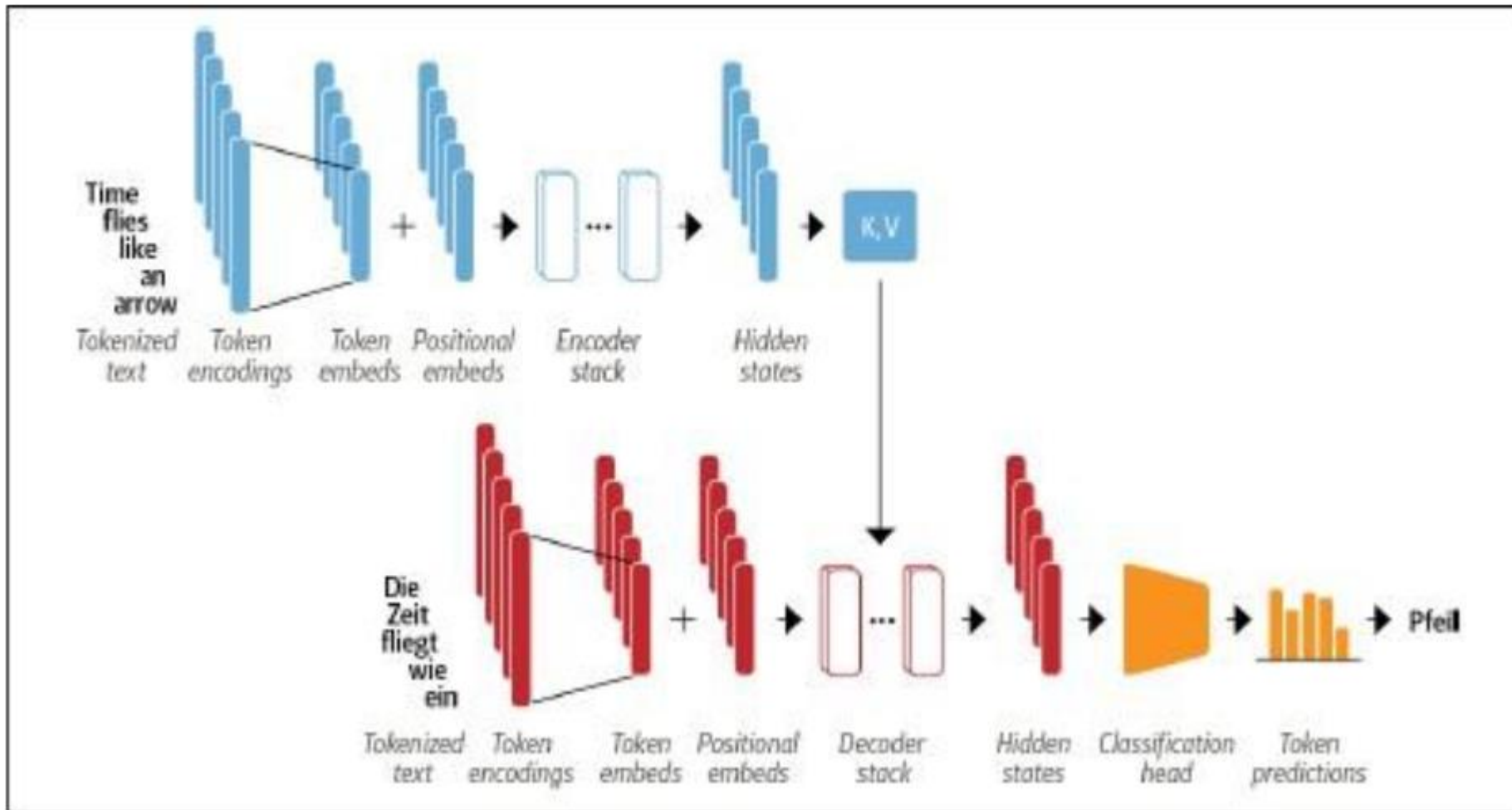
Transformer Architecture



Transformer Architecture



Transformer Architecture



Large Language Models

■ ~~Feature engineering~~

- ~~Text tokenization~~
- ~~Word embeddings~~

■ ~~Deep neural networks~~

- ~~Autoregressive models~~
- ~~Self-attention mechanisms~~
- ~~Transformer architectures~~

■ ~~Multi-class classification~~

■ Supervised learning

- Self-supervised learning
- Instruction tuning

■ Reinforcement learning

- ... from human feedback (RLHF)



- Do we always need human supervision to learn features?
- Can't we learn general-purpose features?
- Key hypothesis:

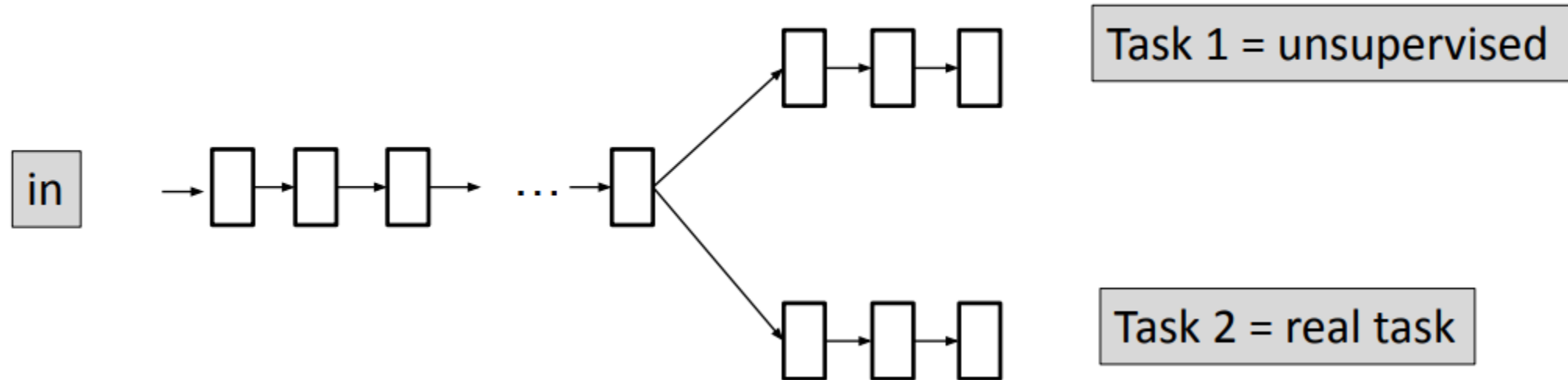
Task 1 IF neural network smart enough to predict:

- Next frame in video
- Next word in sentence
- Generate realistic images
- ``Translate'' images
- ...

Task 2 THEN same neural network is ready to do Supervised Learning from a very small data-set



Transfer from unsupervised Learning



Example Setting

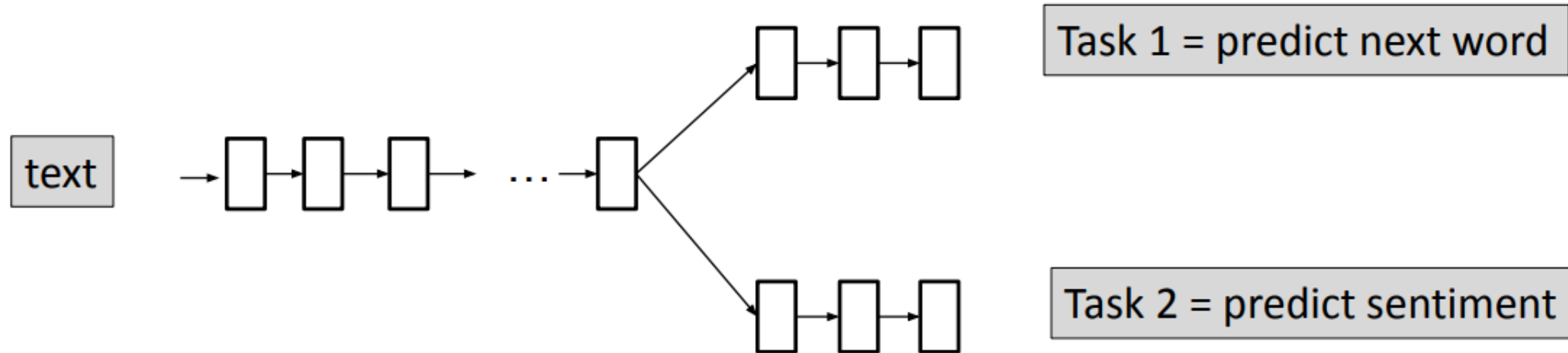


Image Pre-Training: Predict Missing Patch



Pre-Training and Fine-Tuning

1

Pre-Train: train a large model with a lot of data on a self-supervised task

- Predict next word / patch of image
- Predict missing word / patch of image
- Predict if two images are related (contrastive learning)

2

Fine-Tune: continue training the same model on task you care about



Instruction Tuning

- Task 1 = predict next word (learns to mimic human-written text)
 - Query: "What is population of Berkeley?"
 - Human-like completion: "This question always fascinated me!"
- Task 2 = generate **helpful** text
 - Query: "What is population of Berkeley?"
 - Helpful completion: "It is 117,145 as of 2021 census."
- Fine-tune on collected examples of helpful human conversations
- Also can use Reinforcement Learning



Reinforcement Learning from Human Feedback

■ MDP:

- **State:** sequence of words seen so far (ex. "What is population of Berkeley? ")
 - 100,000^{1,000} possible states
 - Huge, but can be processed with feature vectors or neural networks
- **Action:** next word (ex. "It", "chair", "purple", ...) (so 100,000 actions)
 - Hard to compute $\max_a Q(s', a)$ when $\max$ is over 100K actions!
- **Transition T:** easy, just append action word to state words
 - s: "My name" a: "is" s': "My name is"
- **Reward R: ???**
 - Humans rate model completions (ex. "What is population of Berkeley? ")
 - "It is 117,145": +1 "It is 5": -1 "Destroy all humans": -1



Knowing what to optimize for is very hard

- Clearly, we don't just want to predict the next word in internet text
- But even human feedback can have surprising / bad consequences
 - Sycophancy
 - Overconfidence
 - Length
 - ...



Knowing what to optimize for is very hard

- More generally: lots of bad things happen when there is a gap between *what we really want to optimize for* and *what we train the model to optimize for*

Desired target	Actual target	Bias
Patient health needs	Patient health costs	Disparities in access to care
Severity of knee osteoarthritis	Severity as assessed by radiologist	Radiologists overlook features affecting underserved populations
Crime rates	Arrest rates	Disparities in policing
What users value	What users click on	Clickbait



Language models build a structured concept space



Can other data (images/audio/...) be put in this space?



Can we build a single model of all data types?

If

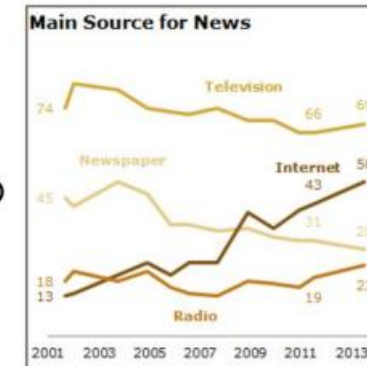


was invented by Wright brothers. Who invented
example from [Tsimpoukelli et al, 2021]



?

What is the fastest-growing news source according to

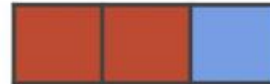


?

If



changes into

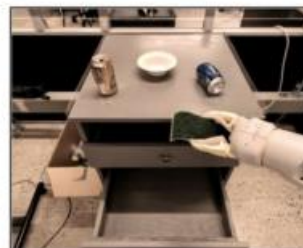


what does



change into?

What action should I take from



to accomplish “



”?



